

Month 2 progress report

Baakhapaa: AI-powered pre-production intelligence system

Weeks 5 to 8 of 12 · Prepared for Baakhapaa · Reference: proposal §8

All four Month 2 deliverables are complete. Each was built ahead of schedule during Month 1 and reported then; this month they were verified in detail, and three of the four were extended in ways the earlier report could not claim. The sections below report against each assigned deliverable in turn.

1. Month 2 deliverables

Wk	Build focus	Deliverable	Status
5	Storyboard generation	Shot types, camera notes, frame controls, regeneration	Done
6	Collaboration and versions	Comment threads, roles, version history with diff	Done
7	Export system	PDF, Word, Final Draft . fdx, production package	Done
8	Subscription tiers	Free / Pro / Studio with checkout and paid-tier gating	Done

Week 5: Storyboard generation

The storyboard engine turns a finalized script into one frame per scene. Each frame is assigned a shot type from a fixed vocabulary of nine — Wide Shot, Medium Wide Shot, Medium Shot, Medium Close Up, Close Up, Extreme Close Up, Over The Shoulder, Point Of View and Insert — chosen from the scene's position within its act rather than at random, so a sequence opens wide and tightens as it approaches its turn.

Each frame also carries a camera note derived from the shot type, the scene's position, its cast, time of day and emotional beat. These are produced without an API call. Before this work, the field was written as an empty string on every frame ever generated.

Frames draw their description from the written draft where one exists and from the structure beat where it does not, and they carry the scene's location, time of day, cast and the project's genre. A writer can override the shot type, edit the camera note, reorder frames and regenerate any single frame. Regeneration re-derives the description from the scene but never overwrites a camera note the writer has edited.

Generation is capped at 24 frames per storyboard, since a long script would otherwise produce a bill proportional to its length. This month the real cost was measured for the first time. Images are produced by gpt-image-1-mini at 1536×1024, and one frame costs **\$0.0033** and takes about 19 seconds.

Item	Previously estimated	Measured
One storyboard frame	\$0.040	\$0.0033
24-frame storyboard	\$0.96	\$0.08
One script with a storyboard	\$1.32	~\$0.44

The earlier estimate assumed DALL-E 3, which the system does not use. The correction matters for pricing: images are roughly 18% of what a script costs to produce and text generation 82%, so the effective control is the length of generated text rather than the frame cap.

Two faults were also found and fixed. The demo placeholder returned an SVG image, which the PDF library cannot embed, so every demo production package printed "frame image not embedded"; the placeholder now requests PNG. And image fetching during export is bounded by a per-fetch timeout and a shared budget, and degrades to a captioned frame box rather than failing, which is the normal case once generated image URLs have expired.

Week 6: Collaboration and version history

Comment threads. Comments anchor to a line of the script and default to the line under the caret, so a note lands where the writer is looking rather than requiring a line number to be typed. Each is attributed to its author and threads are ordered by position in the script rather than by time, so reading the notes means reading the script in order. A viewer may comment; a project administrator may moderate.

Roles. Access is granted per project rather than globally, because a person is usually a writer on their own work and a reader on somebody else's. The three roles are Administrator, Editor and Viewer. The project owner is an administrator implicitly, so no data migration was required to introduce the system. Every protected route takes a minimum required role that defaults to Editor, which means forgetting to mark a route costs a viewer a read and never grants an unintended write.

Sharing is also reachable from inside the editor, not only from settings, since the moment a writer decides to share is the moment they are looking at the work.

Invitations. Anyone can be invited, including a person with no account. The invitation link does not itself grant access — it describes the offer, and membership is granted when somebody registers with the invited address. A link that granted access on click would be a bearer token in a forwarded message. No email is sent, because there is no mail account configured; the inviter passes the link on themselves, which in this market usually means a messaging app rather than email.

Version history. Snapshots are taken automatically on save and coalesced into one per five-minute window, so a long session leaves a readable history rather than hundreds of near-identical entries. A no-op save creates nothing, and a manual save is never merged into an automatic one.

The diff compares any two snapshots and reports ordered diff hunks with line numbers and two lines of context. This replaced a set-based comparison that reported a moved line as no change at all and collapsed every blank line.

Week 7: Export system

Four export formats are available.

- **PDF** for reading and sending. Devanagari now renders correctly: the Noto Sans Devanagari font is bundled with the application under the SIL Open Font License, with its provenance recorded, and takes precedence over the non-redistributable Windows font the system previously fell back to. A test fails if the asset is removed.
- **Word (.docx)** for editing outside the application.
- **Final Draft (.fdx)** for opening in Final Draft, Celtx or Arc Studio. The importer added this month is its exact inverse, so a script can leave the system and return without losing structure.
- **Production package**, a single PDF containing a title page, the screenplay, a shot list carrying slugline, cast, beat, action and camera notes per shot, and a storyboard section with the frames embedded.

Every export is titled and named after the project. All four previously downloaded as `script.pdf` titled "Baakhapaa Script", regardless of which project produced them.

Import was added alongside export this month, which the proposal did not ask for but which the export formats made obvious: a writer arriving with a finished screenplay had to retype it before the system would say anything about it. Final Draft, Fountain, plain text, Word and PDF are all accepted. PDF is the only lossy path, and a scanned page is refused with an explanation rather than imported as an empty script.

Week 8: Subscription tiers

Three tiers are live: Free, Pro at Rs 999 a month and Studio at Rs 2,499.

Payment gateways. Stripe alone could not collect from most Nepali cards, which made the billing system untestable against its actual market. Khalti and eSewa now sit alongside it behind one interface, chosen per checkout. Only Stripe has a subscription primitive: a plan bought through Khalti or eSewa is a single payment that sets an expiry thirty days out and lapses to free when it passes. Every tier check reads through one function, so an expired month reads as free everywhere in the system.

A payment row is written **before** the user leaves for the gateway. This matters because a user returns from Khalti holding only a payment index; if the tier were taken from that returning request, anyone could return claiming Studio. On return the gateway is asked directly what happened, and the amount is checked against the price recorded at the start.

Tier enforcement. Free, Pro and Studio differ in ways the code actually enforces rather than only advertises: three active projects on Free, AI generation and improvement limited to paid tiers, Word and production-package export gated, storyboard generation gated, and collaborator seats limited to two on Free, five on Pro and unlimited on Studio, counted against the project owner's plan. Studio previously had no enforced difference from Pro at all.

Where a free user meets a paid feature, the editor now offers the plan rather than printing the refusal as an error, so the paid tabs are no longer dead ends.

BAAKHAPAA Back to Dashboard

PRICING

Every story deserves a fair start.

Start free. Upgrade when your production does.

PAY WITH

- Khalti**
Sandbox — real gateway, no real money
- eSewa**
Sandbox — real gateway, no real money
- Card (Stripe)**
Set STRIPE_SECRET_KEY (sk_test_...) to reach Stripe's sandbox

Free

Everything you need to write and finish one script

Rs. 0 forever

- ✓ 3 active projects
- ✓ The full course — both tracks, 19 lessons, graded
- ✓ Craft linter: line-by-line notes as you write
- ✓ Craft patterns matched to what is wrong with your scene
- ✓ Benchmark your draft's shape against the corpus
- ✓ Structure starter, scene index, corkboard and outline
- ✓ Version history, diffs and comments
- ✓ Export to PDF and Final Draft (.fpx)

Pro

For writers who want the AI doing the heavy lifting

Rs. 999 per month

MOST POPULAR

- ✓ Everything in Free, with unlimited projects
- ✓ Up to 5 collaborators per project
- ✓ AI scene generation, rewrites and suggestions
- ✓ AI storyboard frames from your scenes
- ✓ Word and production-package exports
- ✓ Structure generation grounded in the craft corpus

Studio

For production teams

Rs. 2,499 per month

- ✓ Everything in Pro
- ✓ Unlimited collaborators per project
- ✓ Roles per project — who can write, who can only read
- ✓ Line-anchored comments, attributed to whoever left them

Figure 1. The three tiers. The pricing page was corrected this month in both directions: Studio had advertised real-time collaboration that was descope, a ten-seat cap that nothing enforced, and priority support with no channel, while the Free tier omitted the course, the craft checker, the corpus benchmark, version history and Final Draft export entirely. Tests now pin the specific untrue sentences.

2. Verification carried out this month

The four deliverables above were re-tested rather than rebuilt. Three activities produced that verification.

Test coverage. Three parts of the backend had no tests at all, and they were three of the riskiest: the payment webhook, which grants a paid subscription and is the only endpoint that does so without a login; the code that decides which fields a user may change; and the retrieval system, which returns an empty result when it fails and therefore reports no error. 104 backend tests were written, and the 26 frontend components without tests were covered. The suite now stands at **765 backend tests in 42 files** and **1,020 frontend tests in 54 files**.

Two security faults surfaced. In the version history, the permission check ran after the other error checks, so the difference between two responses told a logged-in user whether two script versions existed and belonged to the same script, without access to either. Separately the command palette hid itself from logged-out visitors but still called the server; the resulting authentication failure logged them out.

First run with live credentials. Everything to date had been verified against a mock provider. Running against real API keys surfaced the storyboard cost error reported in Week 5, and one fault in the test suite itself: its configuration forces a mock database and skips payment providers, but nobody had done the same for the AI keys, so the tests began calling the live API at cost. A suite that finishes in seconds took 26 minutes.

Writing a screenplay inside the system. A complete short film was written and put through the system's own tools. This found five faults that no test fixture had caught.

1. FADE IN: was counted as a scene, so a 16-scene script reported 17. The error carried into page statistics and into the corpus comparison figures shown to writers.
2. The craft panel analysed the wrong text: choosing a problem category made it examine the category's own description rather than the script, then report the result as though found in the writer's draft.
3. Renaming a scene had no effect on the scene list, so the timeline went on displaying a heading that no longer existed in the script.
4. Focus mode drew a 723-pixel page on a 1,274-pixel screen.
5. The editor URL was labelled as carrying a project identifier but carried a script one.

3. Work beyond Month 2 scope

Three additions were made that the proposal does not schedule. They are recorded here for completeness rather than claimed as deliverables.

Retrieval measurement. The retrieval layer, which supplies craft techniques to the AI prompt, had never been measured. A test set of 34 cases was built and the baseline is **20% precision@1** on the queries the editor actually sends. An initial figure of 82.4% proved misleading: twenty-nine of the cases search using text that was itself used to build the index, which makes a correct answer close to guaranteed, and averaging them with the five realistic cases hid the number that mattered.

Streaming. AI requests previously waited for the entire response before showing anything. Scene generation and rewriting now stream as the text arrives.

Character analysis. A view called Cast gathers each character's dialogue in one place and reports line length, vocabulary repetition, and how often the character asks a question rather than makes a statement. On a test screenplay one character asks in 40% of his lines and another in none, which makes a difference in voice visible as a number.

The screenshot shows the 'Cast' view of a screenplay software. At the top, there's a timeline with scene markers: 1 INT. CHIYA PASAL, PATAN - MORNING, 2 EXT. BAGMATI BRIDGE - DUSK, and 3 INT. CALL CENTRE, NAXAL - NIGHT. The main area is divided into three panels. The left panel shows character statistics for SANJANA and RAAJA. The middle panel displays the script text with line numbers. The right panel contains writing prompts and analysis tools.

Character	Words per Line	Vocabulary	Asks
SANJANA	5.3	0.94	0.33
RAAJA	3.5	1	0

Script text:

INT. CHIYA PASAL, PATAN - MORNING

Steam rises off the kettle. SANJANA wipes the counter without looking at it.

SANJANA

Timro result aayo?

RAAJA

Bholi.

SANJANA

Bholi bhaneko kahile? Timi hardin bholi bhanchhau.

EXT. BAGMATI BRIDGE - DUSK

Raaja waits at the rail, watching the water go under.

RAAJA

Ma pani aaunchu. Tara aile hoina.

INT. CALL CENTRE, NAXAL - NIGHT

Forty people saying the same sentence in different voices.

SANJANA

Figure 2. The Cast view with one character opened. Each line carries its number in the script, and clicking a line moves the cursor to it.

Interface work in the same period moved the corkboard and outline beside the script rather than replacing it, corrected the page height to match the pagination rule, applied standard screenplay margins, and made scene headings renameable and act durations editable directly on the timeline.

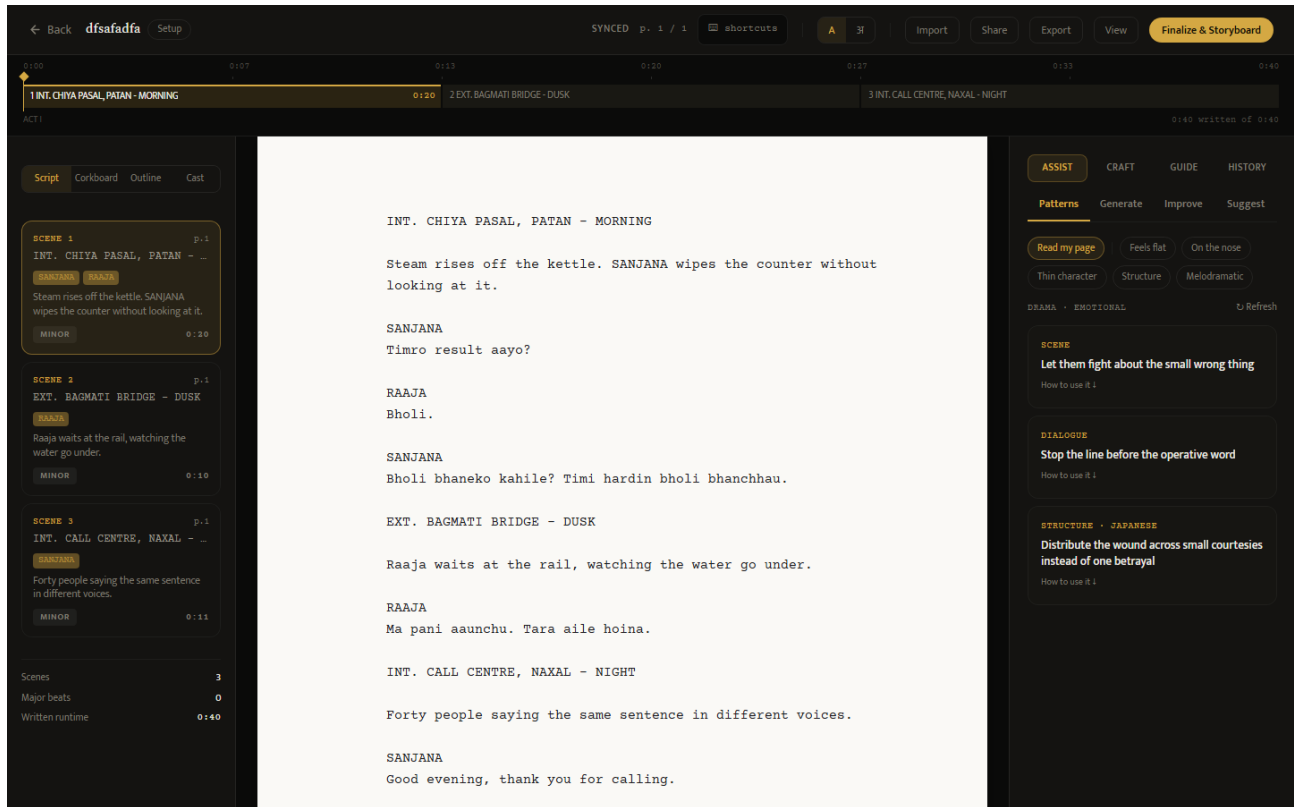


Figure 3. The workspace at the end of Month 2, with the four views and scene index cards on the left, the screenplay page in the middle and the assistant panel on the right.

4. Verification status

Both AI providers can be reached with live credentials, and the image generation path has been run end to end at real cost. Text generation returns a billing error because the account holds no credit, so the paid generation features remain verified against the mock provider only.

The system has not been deployed. It runs on a local database because none has been created in the cloud, which leaves four schema migrations unapplied, and no payment has been processed through any gateway in any of the three currencies the system now supports.

A cloud database, a live server and a first real payment are the work of Month 3. Every remaining estimate depends on them.

Figures were captured from the running system using `baakhapaa-frontend/scripts/capture-screenshots.mjs`.